



Grade 5 Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 5 · FULL YEAR

160 min per week · 2 lessons weekly

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in activities and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade 5 Science is the capstone of elementary science, aligned to both NGSS and Florida's CPALMS benchmarks. The year runs as 32 modules in six units; each module pairs a concept lesson with an investigation, model, data task, or engineering challenge. Scholars plan fair tests with variables and repeated trials, build particle models of matter, wire circuits, trace energy from the Sun through ecosystems, map our place in the Milky Way, and close the year designing a family disaster-preparedness plan.

THE YEAR AT A GLANCE

1 Practice of Science (Modules 1–2)

Observation and evidence, testable questions, variables, control groups, repeated trials, and defending conclusions.

3 Energy, Electricity, and Motion (Modules 10–16)

Forms of energy, transformations, circuits and conductors, static electricity, gravity, and how force and mass change motion.

5 Earth and Space (Modules 24–28)

Galaxies and the Milky Way, star brightness and distance, the solar system, inner and outer planets, and patterns in the sky.

2 Matter and Its Properties (Modules 3–9)

Particles too small to see, states of matter, identifying materials, dissolving, mixtures, conservation of matter, and physical versus chemical changes.

4 Living Organisms and Ecosystems (Modules 17–23)

What plants need to grow, food energy from the Sun, human organs and body systems, structures, adaptations, and matter cycling through ecosystems.

6 Water, Weather, Climate, and Preparedness (Modules 29–32)

The water cycle and the ocean, what determines weather, climate zones, and an engineering challenge: design a family preparedness plan.

A TYPICAL WEEK

- Two science lessons each week, completing one module.
- The first lesson introduces the week's core idea with readings, examples, and guided observation.
- The second is hands-on: an investigation, model, data task, or engineering challenge where scholars collect evidence and defend conclusions.

TIME COMMITMENT

160 minutes / week

two lessons a week, about 80 minutes each

- Each module spans one week: one concept lesson, one investigation.
- Investigations with repeated trials and the preparedness design challenge may run a little longer.

WHAT SCHOLARS NEED

- A science notebook for observations, data tables, graphs, and models.
- Simple household materials for investigations (and basic circuit components when the course calls for them).
- A computer with reliable internet and access to Canvas.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting twice a week on the school's weekly schedule.